

Table 1: Cost analysis comparing KCEA against baselines on seven benchmarks. Each cell reports the mean $\pm$ std across datasets. Best (lowest) per column in **bold**. Alignment time is shown only for KCEA; baselines have no alignment step.

(a) Trainable parameters.

Method	Peak train. (M) ↓	Final (M) ↓	Total (M) ↓	Share (%) ↓
APER+Adapter	—	—	—	—
APER+SSF	—	—	—	—
APER+VPT-Deep	—	—	—	—
CODA-Prompt	—	—	—	—
DUCT	—	—	—	—
DualPrompt	—	—	—	—
EASE	—	—	—	—
L2P	—	—	—	—
MOS	—	—	—	—
SLCA	—	—	—	—
TUNA	—	—	—	—
KCEA	—	—	—	—

(b) Memory cost (MB).

Method	Peak train. RAM (MB) ↓	Final RAM (MB) ↓
APER+Adapter	—	—
APER+SSF	—	—
APER+VPT-Deep	—	—
CODA-Prompt	—	—
DUCT	—	—
DualPrompt	—	—
EASE	—	—
L2P	—	—
MOS	—	—
SLCA	—	—
TUNA	—	—
KCEA	—	—

All values exclude the shared frozen backbone (327 MB, ViT-B/16). For KCEA, final RAM includes class-conditional Gaussian statistics (μ_c, Σ_c).

(c) Wall-clock time (s, cumulative across all tasks).

Method	Training (s) ↓	Evaluation (s)	Alignment (s)
APER+Adapter	—	—	—
APER+SSF	—	—	—
APER+VPT-Deep	—	—	—
CODA-Prompt	—	—	—
DUCT	—	—	—
DualPrompt	—	—	—
EASE	—	—	—
L2P	—	—	—
MOS	—	—	—
SLCA	—	—	—
TUNA	—	—	—
KCEA	—	—	—